

Self Efficacy/Health Behavior Model

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1 SELF EFFICACY

1.A Socioeconomic Status

Socioeconomic status has always had a strong impact on health behaviors across the population. This status influences self-efficacy by determining the type and amount of resources that is available to an individual. People in lower levels of status tend to live in environments in which they face more difficult barriers preventing a healthy lifestyle, such as healthy food and affordable healthcare. Over 90% of U.S. healthcare spending goes toward treating illness rather than prevention, even though half of deaths are preventable (Parker). This ultimately results in weakened self efficacy based on socioeconomic status. These people must deal with poor illness management, delayed healthcare, and increased exposure towards unhealthy factors. On the other hand, the higher you rise within the economic ladder, the more flexibility and access to healthcare you have. The gap between these two groups severely impacts the self efficacy of individuals within the group, creating health disparities where one group is less able to act against illnesses while the other group is consistently having top notch healthcare.

1.B Cultural Values

Cultural values also shape self-efficacy by molding what an individual's behaviors feel possible or acceptable. Healthy behaviors such as exercise and balanced diets are typically regarded as supported throughout most cultures (Balki et al). However, negative aspects of cultures can be seen through the restrictions that are apparent in some traditions. For example, I am an Asian American and I personally believe mental health is a serious topic, but for my family they do not believe mental health is truly as important as made to seems. Therapy is a farce in their belief which made it difficult for me to develop a healthy mental state growing up. In many cultures, similar unhealthy practices such as smoking and drinking are common and sometimes even encouraged. This creates a difficult situation in which a person's self efficacy is damaged and unable to make healthy choices or risk feeling socially isolated. These types of traditions create health disparities because the cultural norm within the community prevents the development of healthy behaviors. Cultural values can force an individual to give up self efficacy in exchange for community involvement, all while building an environment that reinforces unhealthy habits.

2 HEALTH BEHAVIOR THEORY'S APPLICABILITY TO DIGITAL HEALTH DESIGN

2.A Health Belief Model

Personally, I believe that out of all the different theories we have learned so far, the Health Belief Model, or HBM, is the most useful principle that can be used by designers to create digital health interventions that address health disparities. This model gives us a thorough and concrete formula on why people act in specific ways when regarding health behaviors. The theory posits that people act if they believe they are at risk, if the condition is serious, if the benefits outweigh the obstacles, and if they feel capable (Alyafei and Carr). By turning these constructs into design requirements, digital health interventions can be made by designers that target specific health disparities and tailor their designs to overcome disparities.

Firstly, HBM can help intervention designers make changes to designs by making them more tailored to meet the specific needs of individuals. The designers can create tools that better fit people and best assist in developing healthy behaviors. For example, apps can use personalization and customization to show people their risk in a clear way that responds to them best, helping them see why they should care about a condition. Tools that let users see results, such as a continuous glucose monitor, lets the user see immediate results and give the benefits a more sensible understanding. HBM focuses on key features that maximize an individual's response to encourage healthy behaviors. That's why designers add in features such as automatic tracking, simple interfaces, and language options, so that people are more likely to be helped by these technologies.

As someone who is diabetic, I have personally experienced how a change into a different technology can help bridge health disparities. For the past few years, I have used a glucometer to check my blood sugar levels every day. However, I frequently forget and sometimes simply do not want to use the glucometer at all. The process was painful, inconvenient, and required me to remember to do it consistently. According to the Health Belief Model, my perceived barriers were high and the perceived benefits were low. With a change to a continuous glucose monitor, these barriers were essentially completely removed. With a continuous glucose monitor, I no longer needed to remember to take my blood sugar levels every day and the discomfort of the needles was no longer an issue. The machine would automatically give me readings and alerts for my sugar levels and keep track of my condition at all times. The redesign of this technology was able to personally help me improve my health behaviors.

2.B Shortcomings

Although the health belief model is very effective, it does also have its limitations. The main disadvantage that I believe will be difficult to overcome would be the audience in which it's geared towards. HBM is typically meant to focus on individuals instead of large population barriers such as societal and environmental differences (Boskey). For example, an individual may want and try to develop healthy behaviors due to new digital health interventions; however, they may still lack access to safe neighborhoods for exercise and affordable healthy foods. HBM can be effective, but it is unable to account for such situations. Health disparities can be found from a multitude of factors such as location, race, and income. Relying on just the

health belief model is insufficient in being able to eliminate all health disparities. However, it can indeed help even if by a minute amount.

3 HEALTH BEHAVIOR THEORY WITH TECHNOLOGY

3.A MyFitnessPal

MyFitnessPal is an application on your phone that is used to help promote better living and a healthy fitness regimen. The app helps people by tracking their diet, exercise, and physical conditions. Every single day the users are supposed to input what he or she eats for every meal and how much. The application will then calculate the calories and nutrition of the meals and give the user a comprehensive analysis of what he or she ate within a day along with information on how many calories he or she should be eating and what nutrients are needed. The same functionality is done with exercise as well, where the user logs all the exercise done within a day and the app provides recommendations for the user. In addition, it also connects with wearable devices and other fitness apps to help provide the best self monitoring possible. MyFitnessPal aims to make health management as easy and sustainable as possible by being the most thorough and flexible with users.

3.B Health Belief Model

According to the Health Belief Model, MyFitnessPal is very effective in its assistance in healthy behavior by focusing on two different factors, which is reducing perceived barriers and increasing perceived susceptibility. One of MyFitnessPal's main objectives is to have the user log meals within the app every single day to keep track of calories and nutrition (Hahn et al.). Anyone who has done calorie and macro tracking before understands how difficult and inaccurate it can be, especially when working with restaurant bought meals. The app removes this barrier by simplifying the process in which the user is able to scan bar codes for processed foods and inputting food amounts directly into the app. MyFitnessPal would then calculate all the nutritional facts directly for the user without any hassle. The user does not have to worry about calculating grams, breaking down meals, or not being able to log a complicated meal. The app itself deals with the complex barriers that prevents many individuals from accurately tracking nutrition intake.

In addition to reducing perceived barriers, the HBM also indicates that MyFitnessPal also increases perceived susceptibility for the user as well. Once the user has logged all their meals and exercise, the app would then give them a comprehensive analysis of the user's health for the day. The user is also recommended to regularly update the app with his or her weight. This adds to analysis that the app can give. This allows the app to provide caloric intake, weight changes, and potential health problems from poor diet or inactivity. By increasing the user's susceptibility to illnesses by presenting data such as showing how consistently over eating can lead to weight gain, the user can better grasp how likely they are to getting health issues and how they need to change their lifestyle. MyFitnessPal utilizes this HBM construct to connect the users more to their daily actions to best encourage overall better health decisions.

3.C Redesign

Just like any other technology and design, there are numerous ways to improve MyFitnessPal to be even more effective in its promotion of healthy living. Two things from the Health Belief Model that I believe are missing from MyFitnessPal would be better cues to action and better perceived benefits. Currently, MyFitnessPal's cues to action are very limited to simple notifications that remind the user to keep up the tracking of meals and exercise. However, this is a very basic level that will not work for most people. The app needs a more customizable and aggressive response to habit building. I believe giving more specialized notifications such as making sure to drink water, not sitting for too long, and even varying critique of what nutrients you need to eat more of will be more effective in helping the user develop and maintain healthy habits. Users tend to respond more when aspects of the application are more directly personalized to them. By redesigning the notification system to correlate the user's data itself into how notifications are given can greatly increase the app's efficacy. Despite this, two factors that might work against this redesign is the annoyance of increased notifications on the user and the user's mindset on being told what to do. Personally I hate being told exactly what to do and being told what to do multiple times. The sheer amount of notifications can drastically turn people away, especially when they perceive it as an annoyance to their daily routine. Also, being consistently told what to do can also hurt an individual's pride and build resistance against healthy behaviors instead. This gives a difficult task to designers on how to redesign the notification system to be more aggressive, but not too aggressive.

In addition, to improve cues to action, the app desperately needs better perceived benefits. The current version of MyFitnessPal does have a goal tracker for targeted weight; however, that is the only noticeable benefit given to the user. People need to be able to visually see more tangible results and give positive reinforcement to be able to continue difficult tasks. The app needs to include better graphs and create a robust milestone checker with minor goals and prizes for the user to have a stronger perceived benefit from using the app. This will ultimately encourage the user to continue working with the app and eventually build healthy behaviors. However, for this design, two factors that can impede the success would be the complexity of given data, and the balancing of the reward system. Better data and results typically can result in more complex and confusing graphs, so the designer would need to balance out how to keep the design simple so people would be able to understand while also seeing good results. Similarly, the reward system needs to be balanced so that individuals can feel the euphoria from getting a reward but not too much to where it becomes dull. Balancing these concepts is how to best improve this design.

4 REFERENCES

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I have not used generative AI to prepare this assignment.